

Annette Tongsak

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EDUCATION

Oregon State University, Corvallis, OR Anticipated Graduation: June 2026
B.S. Computer Science - Applied Graphics Simulation & Game Programming GPA: 3.93
Relevant Coursework: Computer Graphics & Shaders, Linear Algebra, Parallel Programming, Operating Systems, Analysis of Algorithms, Deep Learning, Machine Learning, Software Engineering, Databases

EXPERIENCE

Crowds Technical Director Intern June 2025 - August 2025
Pixar Animation Studios, Emeryville, CA

- Contributed to production shots in ***Toy Story 5*** by integrating large-scale crowd simulations and character animation assets into the production pipeline
- Developed a Python-based pipeline tool using internal APIs to resolve character pairing issues in crowd simulations, **reducing manual setup time by ~90%** while improving consistency across shots
- Set up, simulated, and debugged large-scale crowd scenes, ensuring stability and compatibility across diverse character assets and simulation configurations
- Created reusable character animation assets and animator-friendly rig controls to support scalable crowd workflows

Pixar Undergraduate Program Technical Director Intern June 2024 - August 2024
Pixar Animation Studios, Emeryville, CA

- Worked within technical pipeline departments (set dressing, layout, modeling, shading, lighting, rigging, and effects) using proprietary and industry-standard tools including Maya, Houdini, Katana, Nuke, RenderMan, and USD
- Collaborated on an animated short, contributing to animation and developing a facial rigging system in Blender to automate texture switching and streamline the animation workflow
- Developed a procedural animation workflow using Houdini (VEX) and Blender to generate character motion from user-defined curves, eliminating manual keyframing

Undergraduate Research Assistant June 2023 - May 2024
Oregon State University, Corvallis, OR

- Research assistant under Dr. Yue Zhang, focusing on computer graphics, data visualization, and machine learning
- Collaborated with a graduate student on a wildlife object detection model commissioned by the Oregon Department of Transportation, contributing to dataset preparation and model development
- Reviewed and synthesized research literature in computer graphics and machine learning, covering topics such as path tracing, multithreading, vectorization, gradient-based optimization, backpropagation, and rendering techniques

PROJECTS

Automatic Lip Sync Tool | *Python, Blender API, WhisperX, Phonemizer* January 2026 - Present

- Developing a Blender add-on that integrates speech recognition and forced-alignment models to generate first-pass lip sync animation within existing character animation workflows
- Evaluated multiple alignment approaches, balancing timing accuracy, runtime performance, and pipeline usability for production-style animation workflows

Random Cobweb Generator | *C++, OpenGL* December 2023

- Developed a procedural cobweb generation system in C++ and OpenGL, inspired by DreamWorks' 2011 paper "Building and Animating Cobwebs for Antique Sets"
- Applied mathematical modeling of catenary curves and parabolas to generate 3D web structures

SKILLS

Programming: Python, C/C++, JavaScript, SQL, OpenGL, GLSL
Graphics, Simulation, & ML: PyTorch, CUDA, OpenCL, OpenMP, MPI
3D & Pipeline Software: Maya, Houdini, Katana, Nuke, Blender, USD, RenderMan
Developer Tools: Git, Perforce, Jira, Visual Studio, VS Code
Platforms: Linux/Unix, macOS, Windows